SYMBIOSIS INSTITUTE OF TECHNOLOGY, PUNE

DEPARTMENT OF ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING

DATA VISUALIZATION LAB

Assignment Practical No. 6

DAX Measures, DAX Studio and Interactive Power BI Dashboard

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Course: Data Visualization Lab | Course Code: TE7663

Aim

To create and validate DAX measures using Power BI and DAX Studio, perform time-based sales analysis, and develop an interactive dashboard using the Adventure Works dataset.

Objectives

- Create key performance indicator (KPI) measures for sales, cost, profit, quantity, orders, customers and profitability.
- Implement at least 20 custom DAX measures using functions such as SUM, AVERAGE, COUNTROWS, DISTINCTCOUNT, CALCULATE, DIVIDE, FILTER and RANKX.
- Analyse monthly sales, previous-period sales, sales variance and sales growth.
- Use DAX Studio to run queries, validate calculations and inspect query performance.
- Build an interactive Power BI dashboard with KPI cards, at least four visualizations, and Year, Product Category and Region slicers.
- Interpret product, category, region, sales-trend and profitability results.

Dataset and Tools

Dataset: Adventure Works dataset, using the combined Sales Data (All Years) table and relevant lookup tables such as Product Lookup, Product Subcategories Lookup, Product Categories Lookup, Customer Lookup, Calendar Lookup and Territory Lookup.

Tools: Power BI Desktop, Power Query Editor (for previously completed preprocessing), DAX, and DAX Studio.

Procedure Followed

Step 1 — Opened the existing Power BI model

Continued with the Adventure Works model and the preprocessing completed earlier. The combined sales table and lookup tables were used for the analysis.

Step 2 — Reviewed the data model

Reviewed the relationships between the sales table and Product, Customer, Calendar and Territory lookup tables. Correct active relationships are necessary for measures and slicers to filter the report correctly.

Step 3 — Created DAX measures

Created measures for sales, cost, profit, quantity, orders, customers, averages and profit margin. Additional measures were created for product counts, row counts, quantity statistics, filtered sales and product ranking.

Step 4 — Created time-based measures

Created measures for previous-month sales, sales variance and sales growth percentage. The calendar date field was used to analyse sales over time.

Step 5 — Built the interactive dashboard

Added KPI cards and charts to compare sales by product category and region, monthly sales trends, top products and profit by category. Added slicers for Year, Product Category and Region.

Step 6 — DAX Studio validation and performance review

DAX Studio is used to execute DAX queries against the model, compare query results with Power BI measures and inspect query performance. Insert the actual query/result evidence and performance observations after completing these checks.

Step 7 — Interpreted results

Reviewed KPI values, category and regional comparisons, top-selling products, monthly trends and profitability. The final findings should be based on the values displayed by the completed report.

DAX Measures and Calculations

The following measures were created for the practical. The formulas assume the listed column names and valid relationships in the model. If the names in the Power BI file differ, use the actual field names from the model.

1. Total Sales

```
Total Sales =
SUMX(
    'Sales Data (All Years)',
    'Sales Data (All Years)'[OrderQuantity]
        * RELATED('Product Lookup'[ProductPrice])
)
```

2. Total Cost

```
Total Cost =
SUMX(
    'Sales Data (All Years)',
```

```
    'Sales Data (All Years)'[OrderQuantity]
        * RELATED('Product Lookup'[ProductCost])
)
```

3. Total Profit

```
Total Profit = [Total Sales] - [Total Cost]
```

4. Total Quantity

```
Total Quantity = SUM('Sales Data (All Years)'[OrderQuantity])
```

5. Total Orders

```
Total Orders = DISTINCTCOUNT('Sales Data (All Years)'[OrderNumber])
```

6. Total Customers

```
Total Customers = DISTINCTCOUNT('Sales Data (All Years)'[CustomerKey])
```

7. Average Sales (average sales per row)

```
Average Sales =
AVERAGEX(
    'Sales Data (All Years)',
    'Sales Data (All Years)'[OrderQuantity]
        * RELATED('Product Lookup'[ProductPrice])
)
```

8. Average Order Value

```
Average Order Value = DIVIDE([Total Sales], [Total Orders], 0)
```

9. Average Product Price

```
Average Product Price = AVERAGE('Product Lookup'[ProductPrice])
```

10. Profit Margin %

```
Profit Margin % = DIVIDE([Total Profit], [Total Sales], 0)
```

11. Total Products Sold

```
Total Products Sold = DISTINCTCOUNT('Sales Data (All Years)'[ProductKey])
```

12. Total Sales Rows

```
Total Sales Rows = COUNTROWS('Sales Data (All Years)')
```

13. Maximum Order Quantity

```
Maximum Order Quantity = MAX('Sales Data (All Years)'[OrderQuantity])
```

14. Minimum Order Quantity

```
Minimum Order Quantity = MIN('Sales Data (All Years)'[OrderQuantity])
```

15. Average Order Quantity

```
Average Order Quantity = AVERAGE('Sales Data (All Years)'[OrderQuantity])
```

16. Highest Product Price

```
Highest Product Price = MAX('Product Lookup'[ProductPrice])
```

17. Lowest Product Price

```
Lowest Product Price = MIN('Product Lookup'[ProductPrice])
```

18. Sales Quantity Above 1

```
Sales Quantity Above 1 =
CALCULATE(
    [Total Sales],
    FILTER('Sales Data (All Years)', 'Sales Data (All Years)'[OrderQuantity] > 1)
)
```

19. Sales Quantity Equal 1

```
Sales Quantity Equal 1 =
CALCULATE(
    [Total Sales],
    FILTER('Sales Data (All Years)', 'Sales Data (All Years)'[OrderQuantity] = 1)
)
```

20. Product Sales Rank

```
Product Sales Rank =
RANKX(
    ALL('Product Lookup'[ProductName]),
    [Total Sales],
    ,
    DESC,
    DENSE
)
```

21. Sales Previous Month

```
Sales Previous Month =
CALCULATE(
    [Total Sales],
    DATEADD('Calendar Lookup'[Date], -1, MONTH)
)
```

22. Sales Variance

```
Sales Variance = [Total Sales] - [Sales Previous Month]
```

23. Sales Growth %

```
Sales Growth % = DIVIDE([Sales Variance], [Sales Previous Month], 0)
```

Formatting note: format Profit Margin % and Sales Growth % as Percentage in Power BI. Do not multiply the ratio by 100 if using percentage formatting. Format sales, cost, profit and average order value as currency where appropriate.

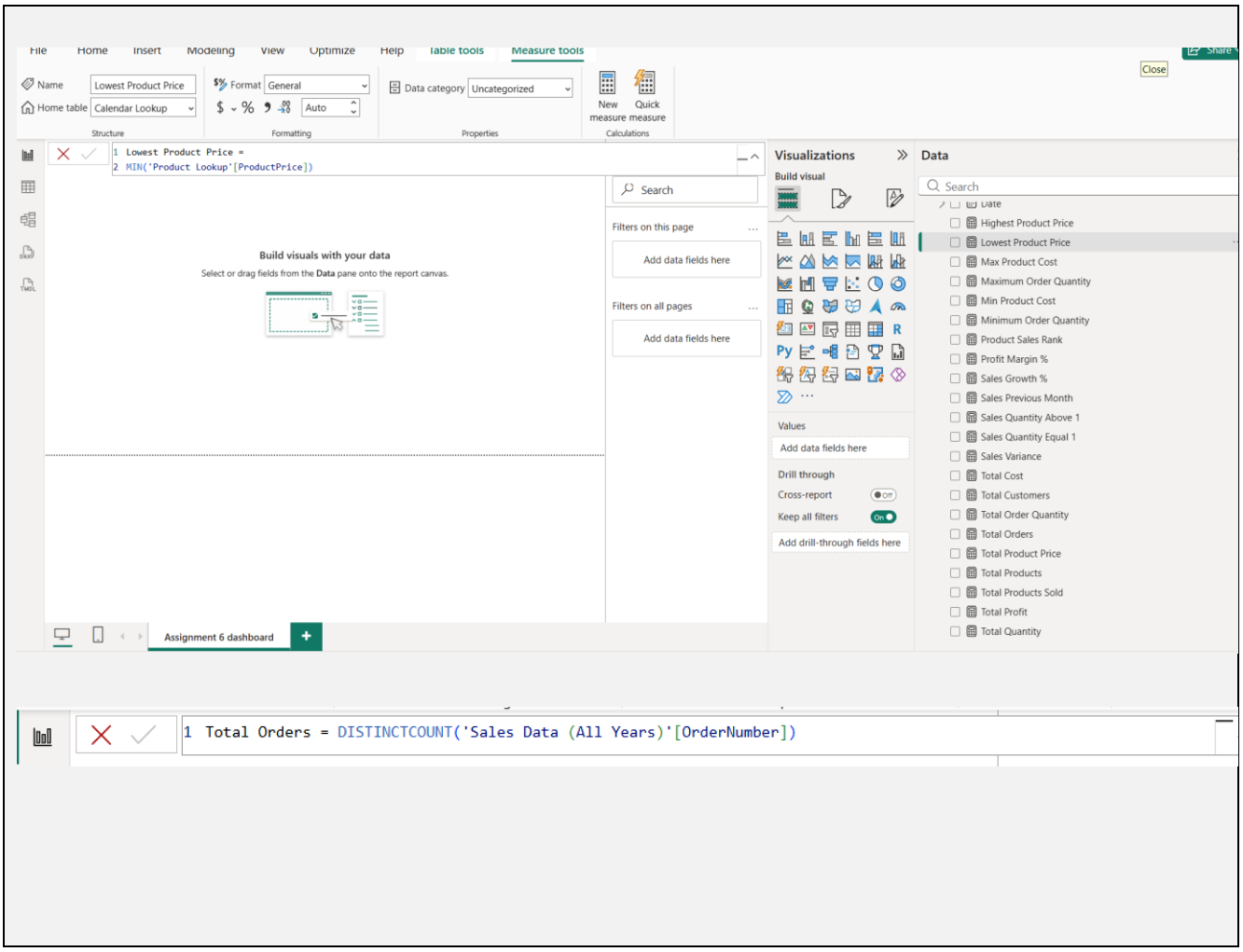
Monthly sales can be shown by placing the Calendar date hierarchy (Year and Month) on a chart axis and [Total Sales] in Values. The [Total Sales] measure responds to the date filter context.

Dashboard Design and Field Mapping

Visual	Fields to use	Purpose
KPI Cards	[Total Sales], [Total Profit], [Total Orders], [Total Customers], [Average Order Value], [Profit Margin %]	Summarise the main business KPIs.
Clustered bar chart	Axis: Product Categories Lookup[CategoryName]; Values: [Total Sales]	Compare sales by product category.
Line chart	X-axis: Calendar Lookup[Date] (Year/Month); Y-axis: [Total Sales]	Show monthly sales trends.
Clustered bar chart	Axis: Territory Lookup[Region]; Values: [Total Sales]	Compare sales by region.

Top 10 bar chart	Axis: Product Lookup[ProductName]; Values: [Total Sales]; visual filter Top N = 10 by [Total Sales]	Identify top-selling products.
Column chart	Axis: Product Categories Lookup[CategoryName]; Values: [Total Profit]	Compare profitability by category.
Slicer — Year	Calendar Lookup[Date] → Year	Filter the report by year.
Slicer — Product Category	Product Categories Lookup[CategoryName]	Filter the report by category.
Slicer — Region	Territory Lookup[Region]	Filter the report by region.

Steps Performed / DAX Studio Evidence



DAX Studio Query and Performance Notes

Query / validation performed:

DAX measures were created in Power BI to calculate Total Sales, Total Profit, Total Orders, Total Customers, Average Order Value, and Profit Margin. The measures were displayed on KPI cards and used in the dashboard charts to analyse sales by product category, year, and region.

Performance observation:

The dashboard displayed the KPI values and visualizations successfully. Total Sales, Total Profit, Total Orders, Total Customers, and Average Order Value were visible on the KPI cards. Actual DAX Studio query execution results and performance timings were not recorded in the provided screenshot.

Dashboard Output Through Cards



Analysis and Interpretation of Results

KPI summary

Total Sales: 24.91M
Total Profit: 10.46M
Total Orders: 25K
Total Customers: 17K
Average Order Value: 990.09
Profit Margin: 0.42 (approximately 42% when formatted as a percentage)

Top-selling products

The provided dashboard does not display a Top 10 Products chart. Therefore, the top-selling individual products cannot be identified from the current screenshot.

Category performance

The Sales by Category chart shows Components as the highest-sales category among the categories visible, followed by Bikes and Clothing. The Total Profit by Category chart also shows Components with the highest displayed profit, followed by Bikes and Clothing.

Regional performance

The Sales by Region chart shows Australia as the highest-sales region among the visible regions, followed by Southwest and Northwest. Australia has the longest displayed sales bar in the current dashboard.

Time-based trends

The yearly sales trend shows an increase from 2020 to 2021, followed by a slight decrease in 2022. Sales reached their highest displayed level around 2021. The screenshot shows yearly data rather than monthly data, so the exact highest and lowest months cannot be identified.

Profitability

The dashboard displays Total Profit of 10.46M and Total Sales of 24.91M. The displayed Profit Margin value is 0.42, which corresponds to approximately 42% when interpreted as a decimal ratio and formatted as a percentage. This indicates that the reported profit is substantial relative to total sales.

Conclusion

This practical demonstrates the use of DAX measures for calculating business KPIs and analysing sales performance in Power BI. The interactive dashboard presents key metrics through KPI cards and visualizations for sales by product category, total profit by product category, yearly sales trends, and sales by region. The displayed results help compare category performance, understand changes in yearly sales, and identify regional differences. The dashboard provides a consolidated view of sales, profit, orders, customers, average order value, and profit margin.